

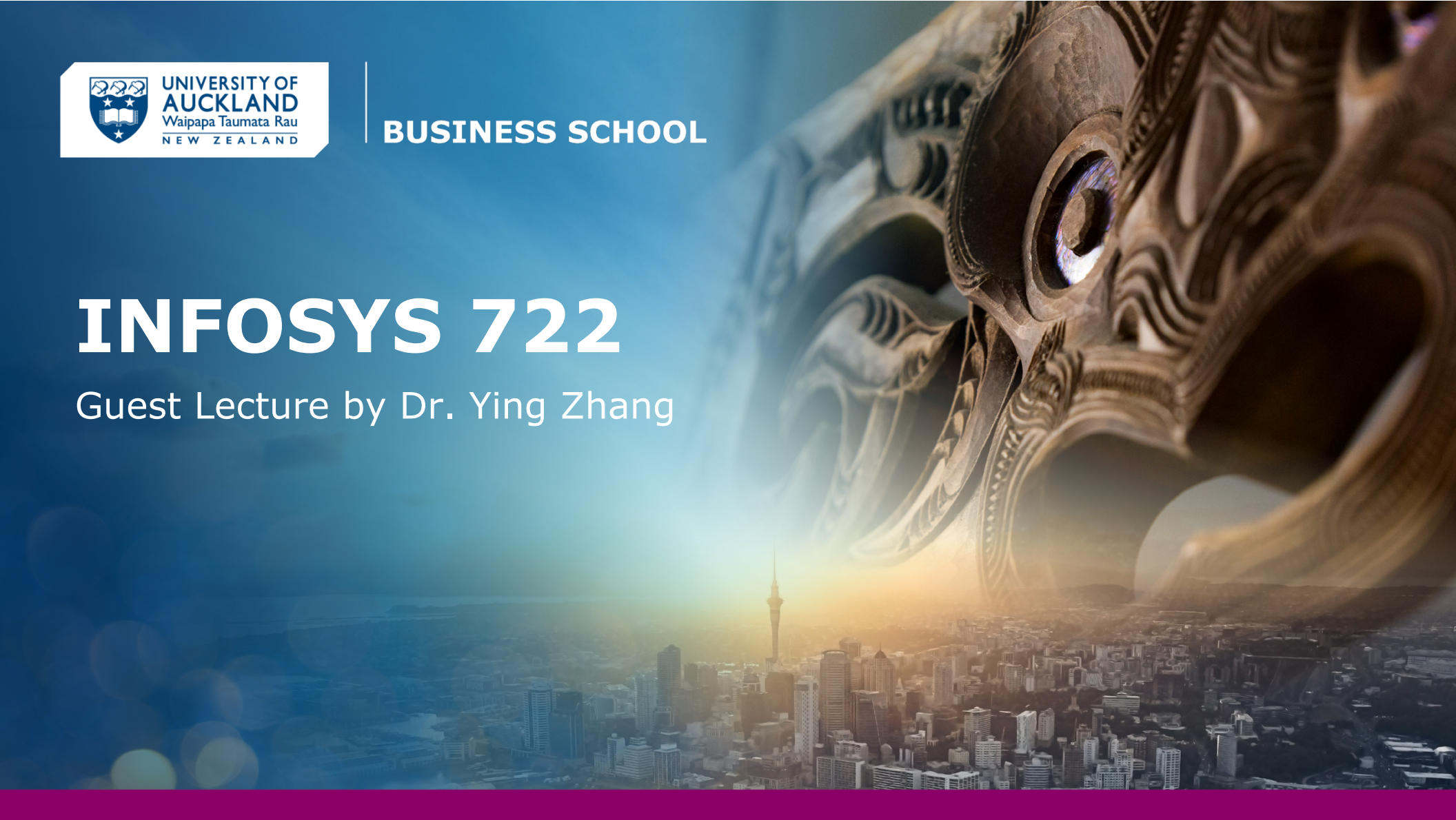


UNIVERSITY OF
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INFOSYS 722

Guest Lecture by Dr. Ying Zhang



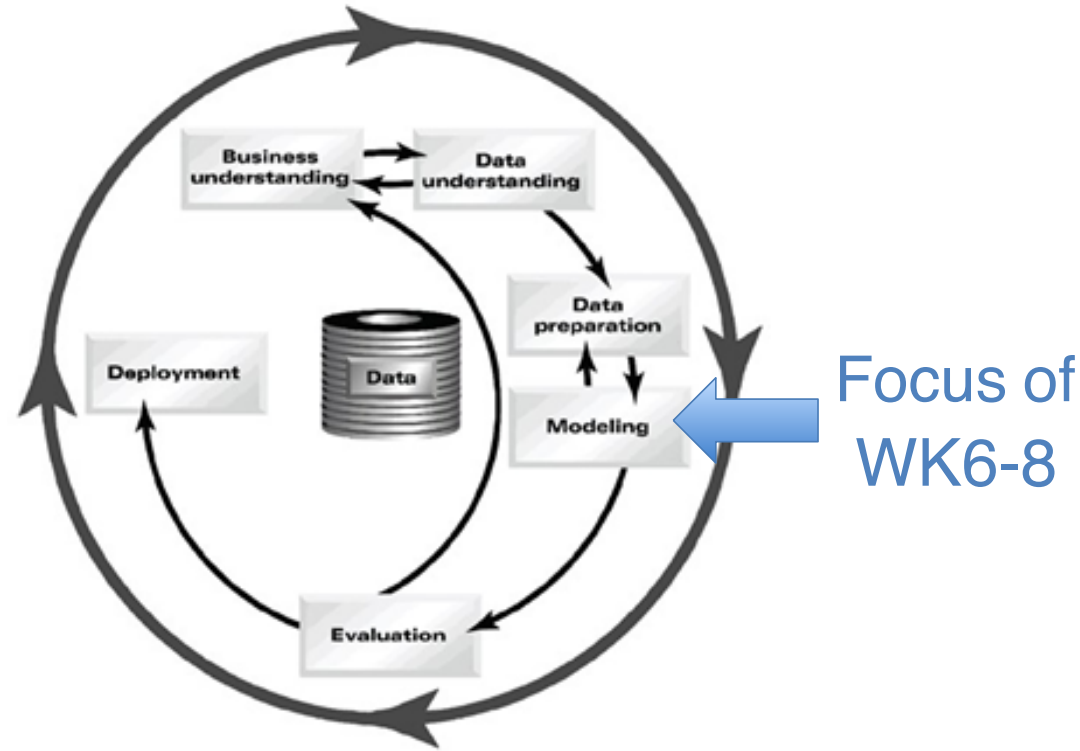
About me

- Dr. Ying ZHANG
 - Bachelor of Computing (First Class Honors), National University of Singapore
 - Ph.D. in Information Systems and Analytics, National University of Singapore
 - Research interests: digital marketing, e-commerce, online word-of-mouth, sharing economy
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Recap

Data mining

- The overall process consisting of business understanding, data gathering and analysis, development of learning models, and adoption of practical decisions and consequent actions based on the knowledge acquired



Use **algorithms** to **discover patterns** in the **data**

Modelling

- In what way can the data be utilised and visualised to answer the question?

Evaluation

- Does the model used perform well or does it need to be adjusted?

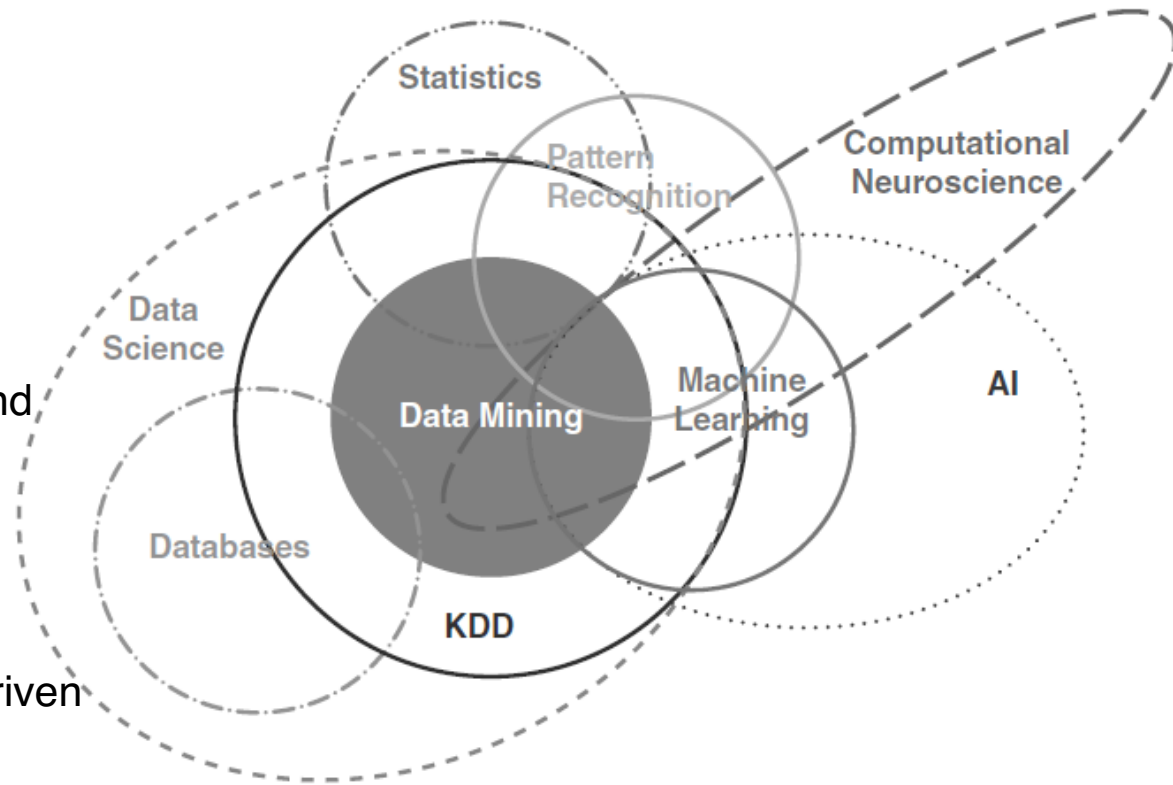
Recap cont.

Multidisciplinary nature of data mining

- Algorithms and models from statistics and machine learning

Data modelling

- Descriptive: summary statistics etc.
- Predictive: statistics/machine-learning driven



Use **algorithms** to **discover patterns** in the **data**



Machine Learning (Not Memorizing)

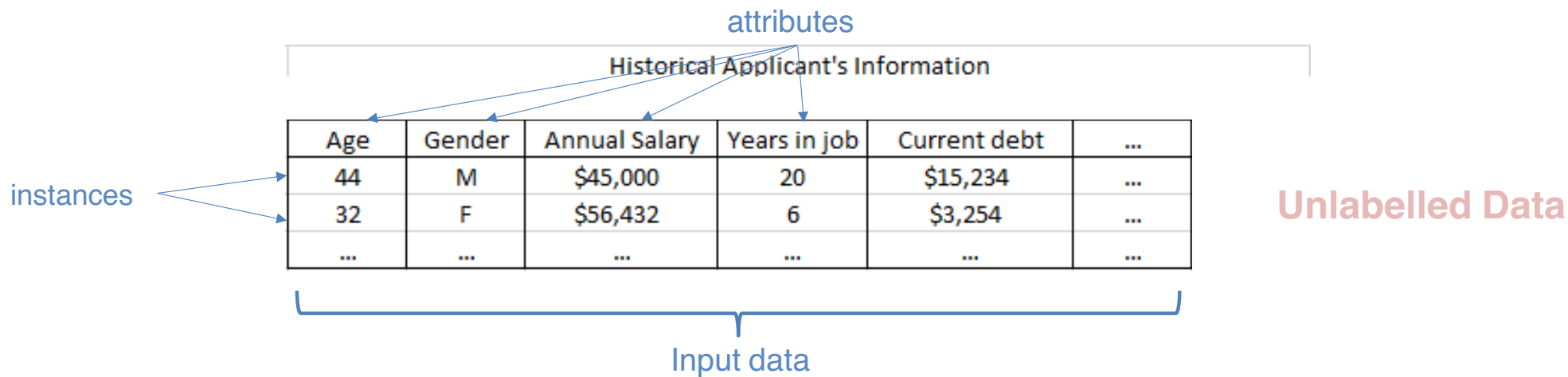
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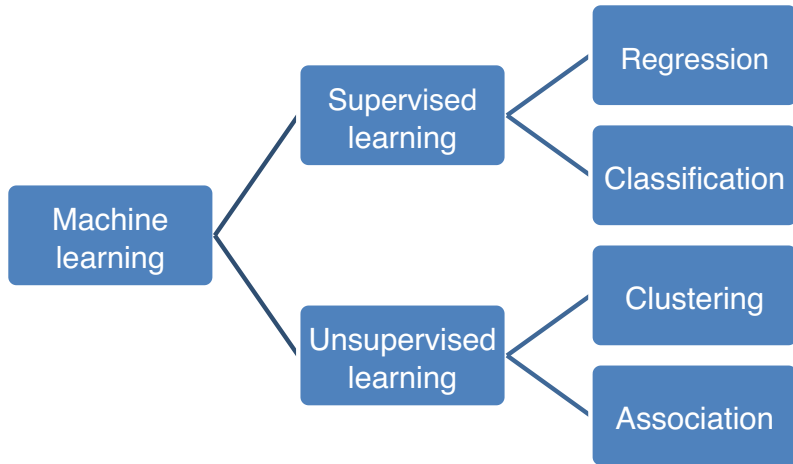
Recap cont.

Basic concepts

- Labelled and unlabelled data



Recap cont.



Supervised learning

- What is known:
 - Given dataset (input)
 - Correct output (labelled data!)
- Having the ideal that there is a relationship between the input and output
- **Regression**: predict results within a continuous output
- **Classification**: predict results with a categorical output

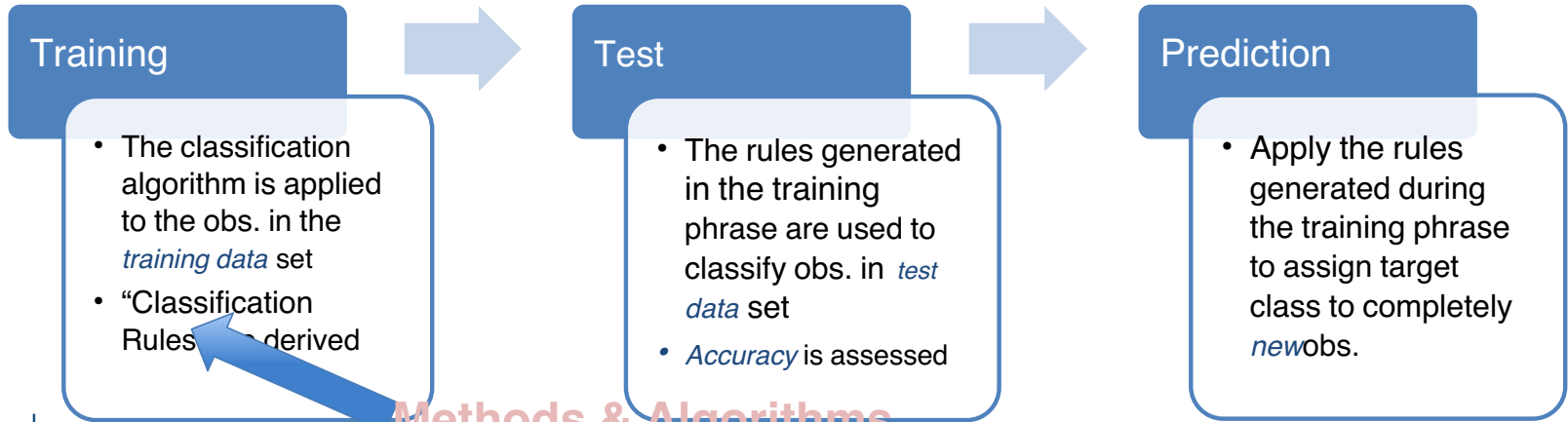
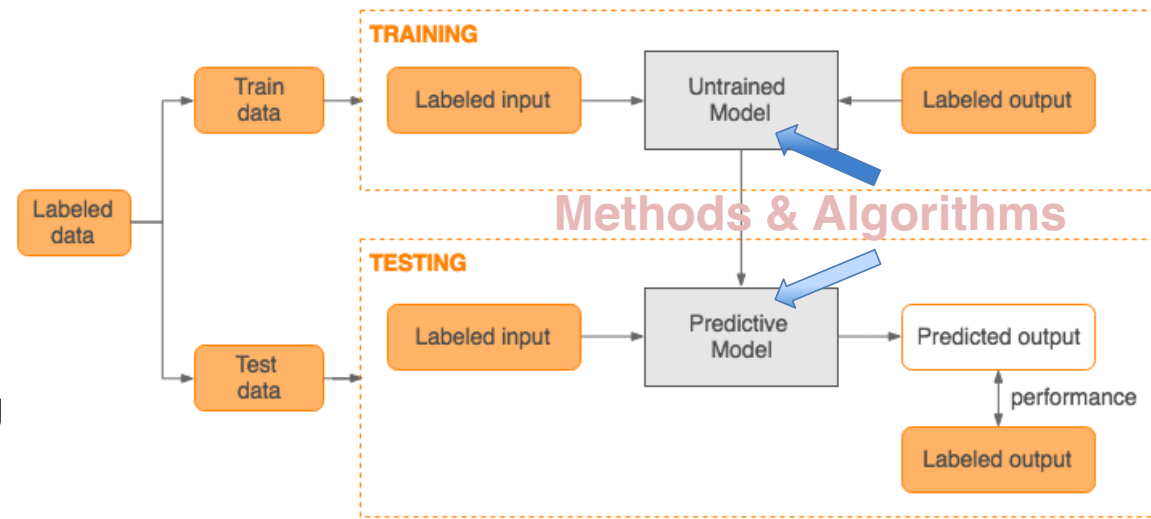
Unsupervised learning

- What is known:
 - Given dataset (input)
- Have little or no idea what our results look like  no feedback based on the prediction results
- **Clustering**: group/cluster data

Recap cont.

Classification Model

- *Supervised* learning methods for predicting value of a categorical target attribute
- Three phrases



Supplementary Videos

- Lecture recording 1: regression, classification
- Lecture recording 2: classification, clustering
- Lecture recording 3: association rules, more about unstructured data

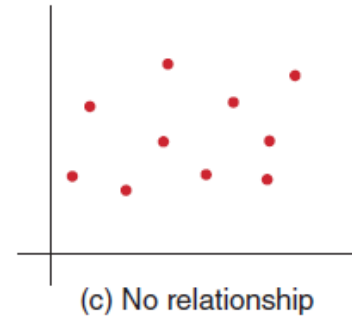
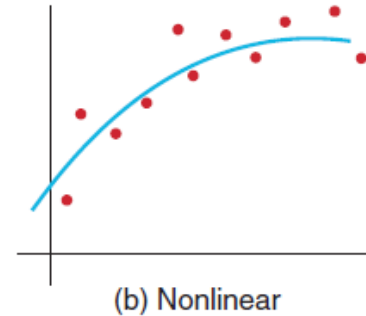
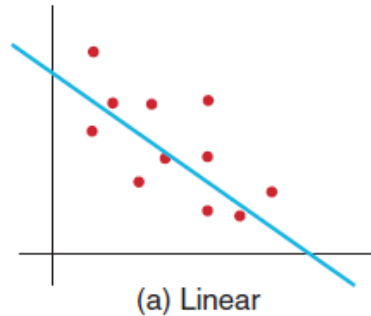
Regression

Regression analysis

- Likely the first ***predictive modelling*** method you learned in data mining
- A tool to build statistical models that model relationships between a dependent variable and one or more independent variables

Let's start with a simple case:

- One dependent variable: continuous
- One independent variable: continuous



How to fit a line?

Linear model

Example: predict weight of a student

- A sample of 19 middle school students, their weight and height

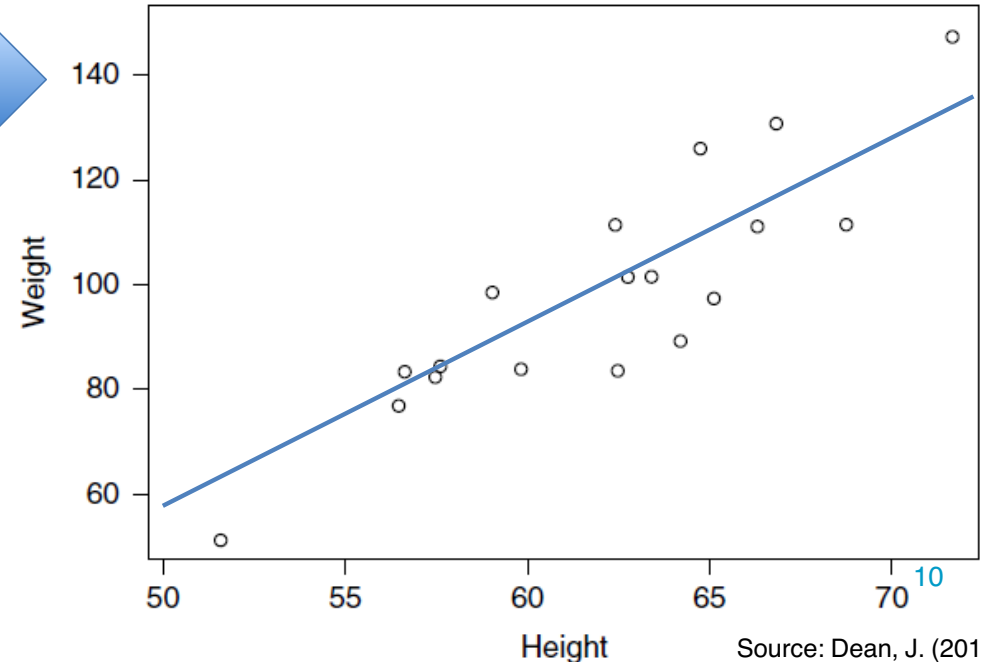
Data understanding & processing

- Summary statistics
- Exploratory data analysis

Let's first consider the case of a “linear line”



Row	Height (inch)	Weight (lb.)
1	75	142
2	60	83
.....		
19	65	98



Source: Dean, J. (2014)

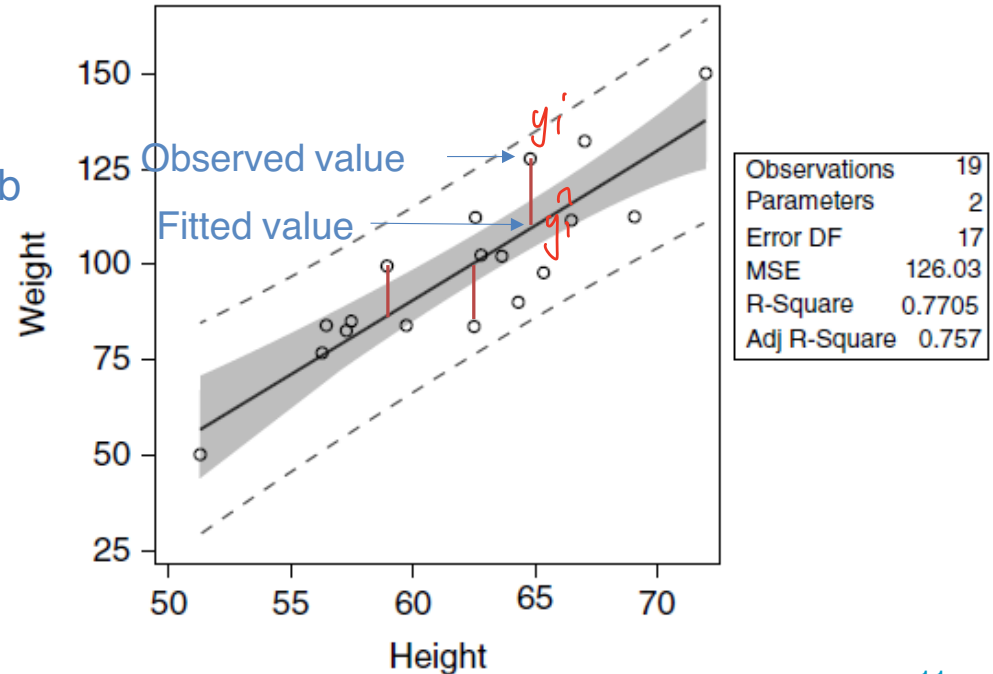
OLS

Best line: minimise the sum of error across all data points

Various methods of linear models

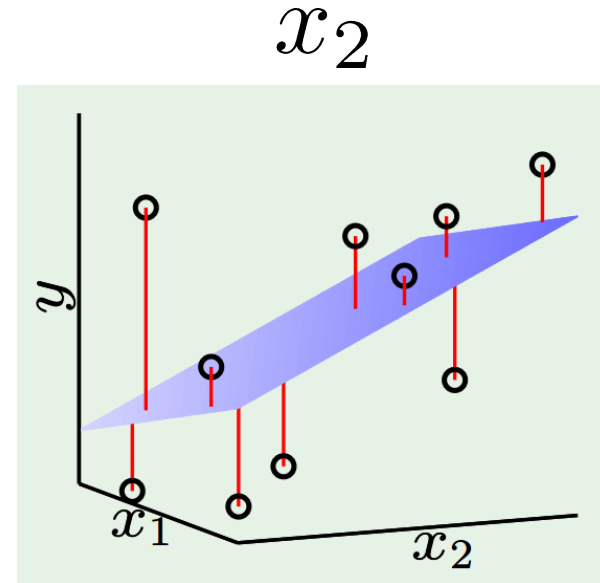
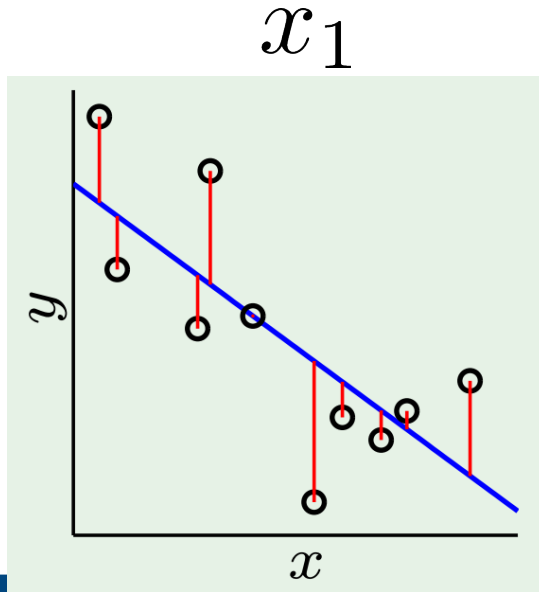
- Ordinary least squares (OLS) ← Week 4's lab
- Minimise the sum of squared residuals (RSS) between the observed value and the predicted/fitted value

$$RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$



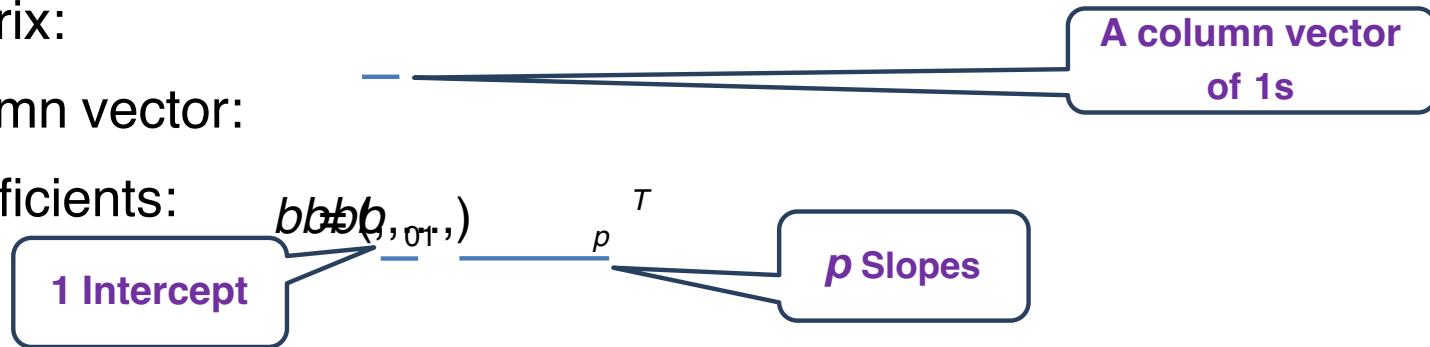
Linear model – more predictors

How about more independent variables?



Linear model – general spec

- Linear regression model of N ($i=1, \dots, N$) observations and p ($j=1, \dots, p$) predictors:
- Where feature matrix:
- Feature x_j is a column vector:
- Parameters or coefficients:



To get parameter values:
OLS

Linear model – estimation

- Minimize the loss function:
- Usual way of solving optimization problems:

Gradient Descent

(For Minimizing)

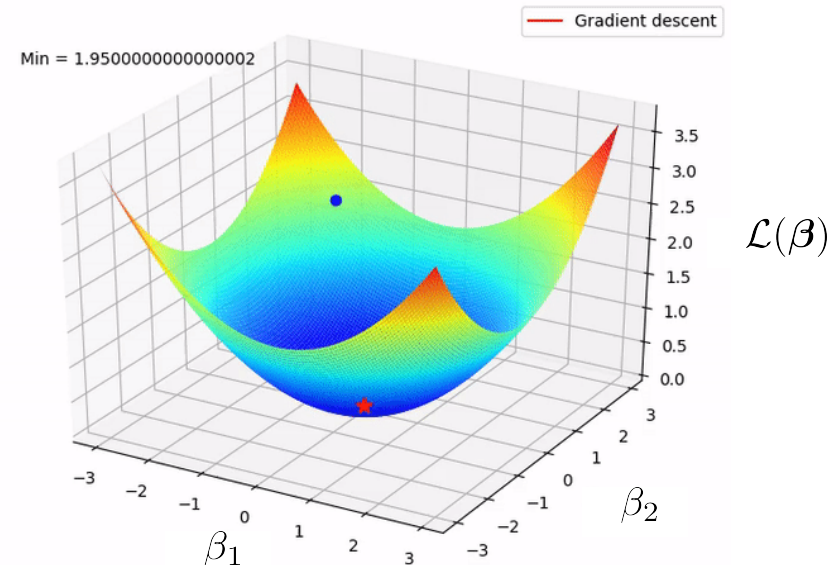


Gradient Ascent

(For Maximizing)



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OLS closed form

- Minimize the loss function:
- Linear model has an even simpler way – a closed-form solution!
- Let the first-order derivative equal to 0 (Normal equations):

$$\frac{\partial}{\partial \beta} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 = 0$$

- To derive this formula:

OLS model fit: R-squared

- R-squared: *Coefficient of determination*
 - The proportion of variation in the dependent variable that is explained by the independent variable of the regression model
 - Total sum of squares (TSS): proportion to the variance of the data
 - Explained sum of squares (ESS)
 - Residual sum of squares (RSS)

R-squared:	0.547
Adj. R-squared:	0.547

OLS – check assumptions

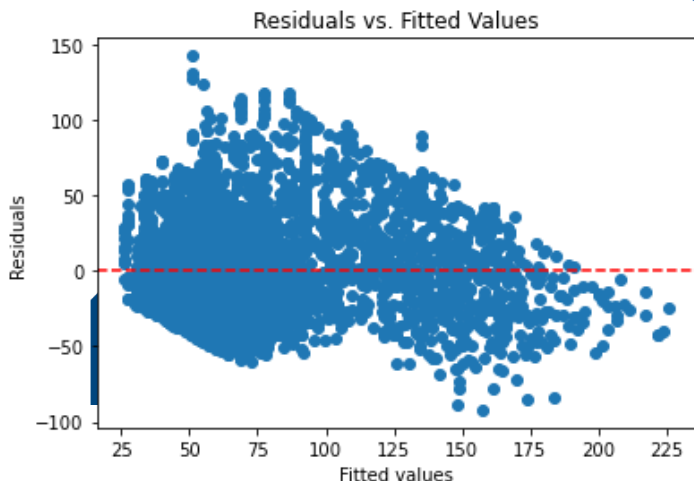
- Linearity
- Independence
- Homoscedasticity (constant variance of errors)
- Normality of errors

```
from statsmodels.stats.diagnostic import het_breuschpagan
```

```
Breusch-Pagan test LM statistic: 1705.952902108297  
Breusch-Pagan test LM p-value: 0.0  
Breusch-Pagan test F-statistic: 198.67043768040304  
Breusch-Pagan test F-statistic p-value: 0.0
```

```
t_breuschpagan(residuals, X1)
```

```
print("Breusch-Pagan test LM statistic:", lm)  
print("Breusch-Pagan test LM p-value:", lm_p_value)  
print("Breusch-Pagan test F-statistic:", fvalue)  
print("Breusch-Pagan test F-statistic p-value:", f_p_value)
```



Omnibus:	475.048	Durbin-Watson:	1.983
Prob(Omnibus):	0.000	Jarque-Bera (JB):	626.624
Skew:	0.805	Prob(JB):	8.52e-137
Kurtosis:	3.647	Cond. No.	492.

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In the big data era ...

Factors to consider for large dataset:

- Computational Complexity
- Storage complexity
- Scalability

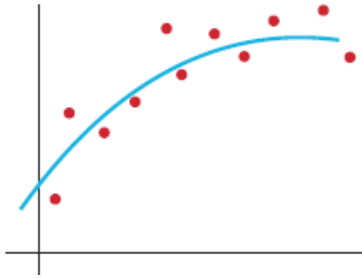
Think about the case that you have 10000000
rows ...



requires $O(nd^2 + d^3)$ operations

- Matrix multiplication
- Matrix inversion
- Matrix vector multiplication

Least squares



Other non-linear function forms

Nonlinear least squares (NLS)

- Minimise minimize the sum of the squared residuals between the observed values and the predicted values

Solved using iterative
optimization algorithms

Classification

What if the dependent variable is discrete or categorical?

Example: predict a simple binary outcome - whether to buy a house or not

- Collect data from $N=1000$ people ($i=1, \dots, 1000$)

Row	Size (m ²)	#Bedroom	#Bathroom	Price (k)	Buy
1	65	2	2	900	1
2	72	3	2	1200	0
.....					
1000	50	1	1	500	0

What happens if we
fit a linear model?

GLM

Generalised linear models (GLM)

- Lift some restrictions, which were prohibitive for certain types of problems
- Offer the flexibility to accommodate response variables that are nonnormally distributed, have a mean with a restricted range, and/or have nonconstant variance— which are all violations of the assumptions for OLS regression

Two common GLM models for binary classification

- Logistic regression model
- Probit regression model



[More in Lecture Recordings](#)

Naïve bayes

- Naïve

- The presence of one feature does not affect the presence of another feature [?] independence of the occurrence of features

- Bayes

- Apply Bayes' theorem to perform probabilistic classification
- Bayes' theorem:
- [?]
 - Show how to update the probability of hypothesis (i.e., proposed model/function) H given new evidence or data E

Naïve Bayes Classification

Given a set of k mutually exclusive and exhaustive classifications $c_1, c_2, \dots, c_k$, which have prior probabilities $P(c_1), P(c_2), \dots, P(c_k)$, respectively, and n attributes $a_1, a_2, \dots, a_n$ which for a given instance have values $v_1, v_2, \dots, v_n$ respectively, the posterior probability of class c_i occurring for the specified instance can be shown to be proportional to

$$P(c_i) \times P(a_1 = v_1 \text{ and } a_2 = v_2 \dots \text{ and } a_n = v_n \mid c_i)$$

Making the assumption that the attributes are independent, the value of this expression can be calculated using the product

$$P(c_i) \times P(a_1 = v_1 \mid c_i) \times P(a_2 = v_2 \mid c_i) \times \dots \times P(a_n = v_n \mid c_i)$$

We calculate this product for each value of i from 1 to k and choose the classification that has the largest value.

Naïve bayes algorithm

- Two classes: "Spam" vs. "Not Spam"
- Training data: labelled emails
- Consider a new text: "Get a free account", to be classified
- One simple idea: compute the probability of spam vs. not spam conditional on the data evidence ?
Choose the classification of the higher probability

- Use Bayes:

$$\text{Spam: } \left(\frac{2}{33} \times \frac{3}{32} \times \frac{2}{33} \times \frac{1}{33} \right) \frac{3}{5}$$

$$\text{not spam} = \left(\frac{0}{9} \times \frac{0}{9} \times \frac{0}{9} \times \frac{1}{9} \right) \frac{2}{5}$$

↓

$$\frac{1}{18} \times \frac{1}{18} \times \frac{1}{18} \times \frac{2}{18}$$

Text	Label
Get a free luxury bag.	Spam
Send me the document.	Not Spam
Buy me a gift card!	Spam
Give me some money.	Spam
Thank you for the update.	Not Spam

???

Naïve bayes algorithm cont.

- Bring in Naïve: every word in the sentence is independent of each other
- Features: words in an email (e.g., “free”, “gift”, “money”) and their word frequencies
- We should be able to calculate all probabilities because each individual word actually shows up several times in the training data
- Apply additive smoothing
 - Add 1 to every count so it is never **0**
 - Meanwhile, add the number of possible words to the divisor, so the division will never exceed 1

Naïve bayes cont.

- Advanced techniques:
 - Removing stop words, lemmatization, more counting techniques (n-grams, tf-idf, ...)
- Advantages:
 - One of the most popular choices for text classification problems
 - Can perform well in multi-class classifications
 - Fast and easy
- Disadvantages:
 - Assumption on independence  cannot learn relationships among features

Nearest neighbour

- Mainly used in the case when all attributes are continuous (but can be modified to deal with categorical attributes)
- Basic k -Nearest Neighbors (k -NN) Classification
 - Find the k training cases that are *closest* to the unseen case
 - Take the most commonly occurring classification for these k cases
- Example:

a	b	c	d	e	f	Class
yes	no	no	6.4	8.3	low	negative
yes	yes	yes	18.2	4.7	high	positive

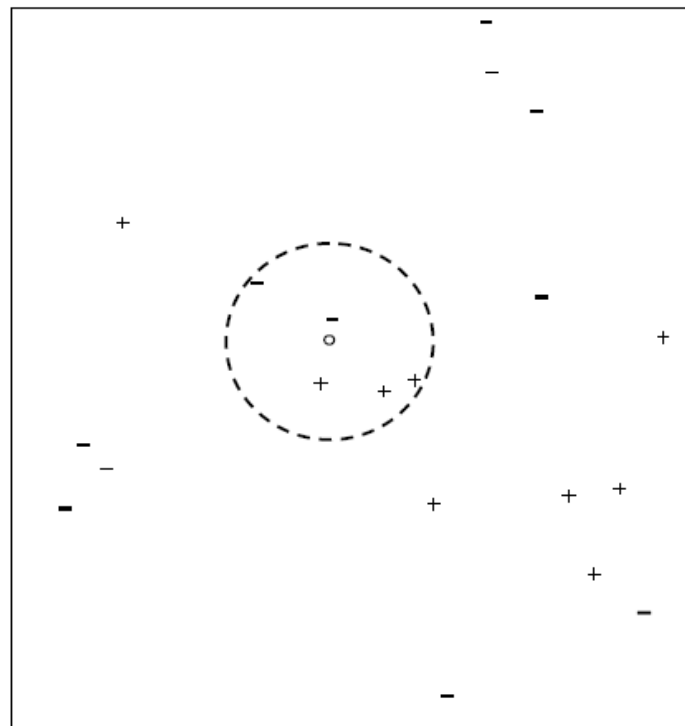
yes	no	no	6.6	8.0	low	????
-----	----	----	-----	-----	-----	------

Nearest neighbour algorithm

Attribute 1	Attribute 2	Class
0.8	6.3	—
1.4	8.1	—
2.1	7.4	—
2.6	14.3	+
6.8	12.6	—
8.8	9.8	+
9.2	11.6	—
10.8	9.6	+
11.8	9.9	+
12.4	6.5	+
12.8	1.1	—
14.0	19.9	—
14.2	18.5	—
15.6	17.4	—
15.8	12.2	—
16.6	6.7	+
17.4	4.5	+
18.2	6.9	+
19.0	3.4	—
19.6	11.1	+

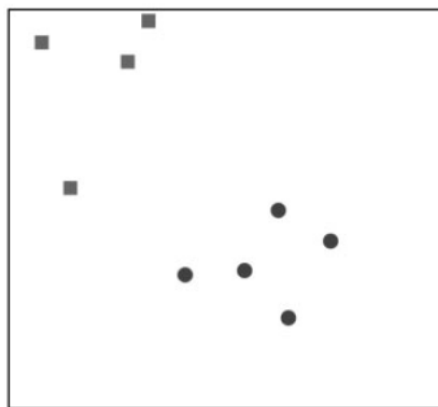
Basic k -Nearest Neighbour Classification Algorithm

- Find the k training instances that are closest to the unseen instance.
- Take the most commonly occurring classification for these k instances.

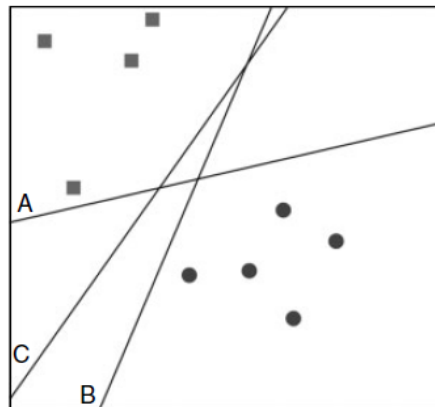
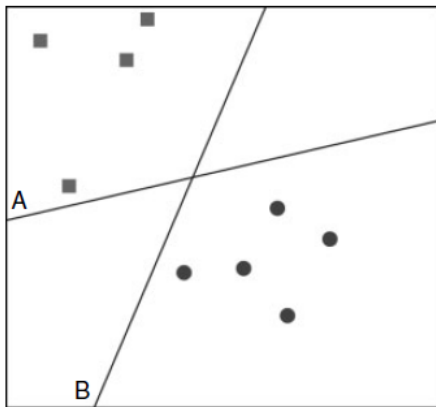


Support vector machine (SVM)

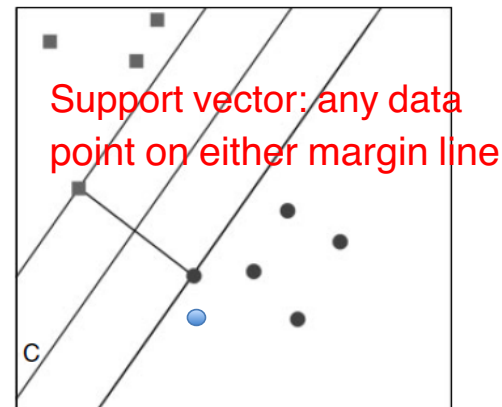
- LDA: linear decision boundaries
- SVM: produces nonlinear boundaries by constructing a linear boundary in a large, transformed version of the feature space
- The support vector classifier (SVC)
 - Maximise the margin (distance) between the nearest data points (support vectors) of different classes



BU



How many for this case?



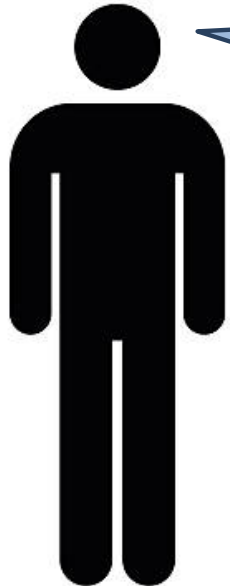
Classification cont.

- So far, classification based on probabilities
- In our daily life, do we always consider conditions and make decisions based on probabilities?

Decision process

Marriage Interview (or “Blind Date”)

Mr. Z

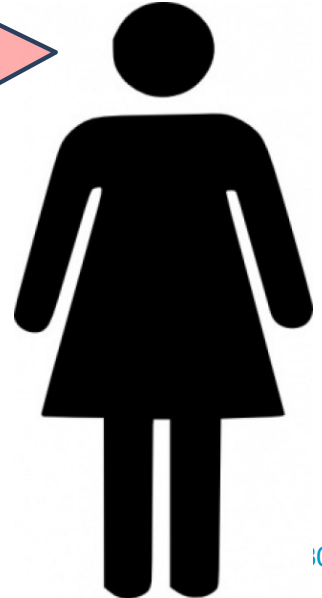


I don't know, think
just ask your son that
official, tax department

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Useless information, this guy is
tricky, I need to be more creative

Mrs. Y



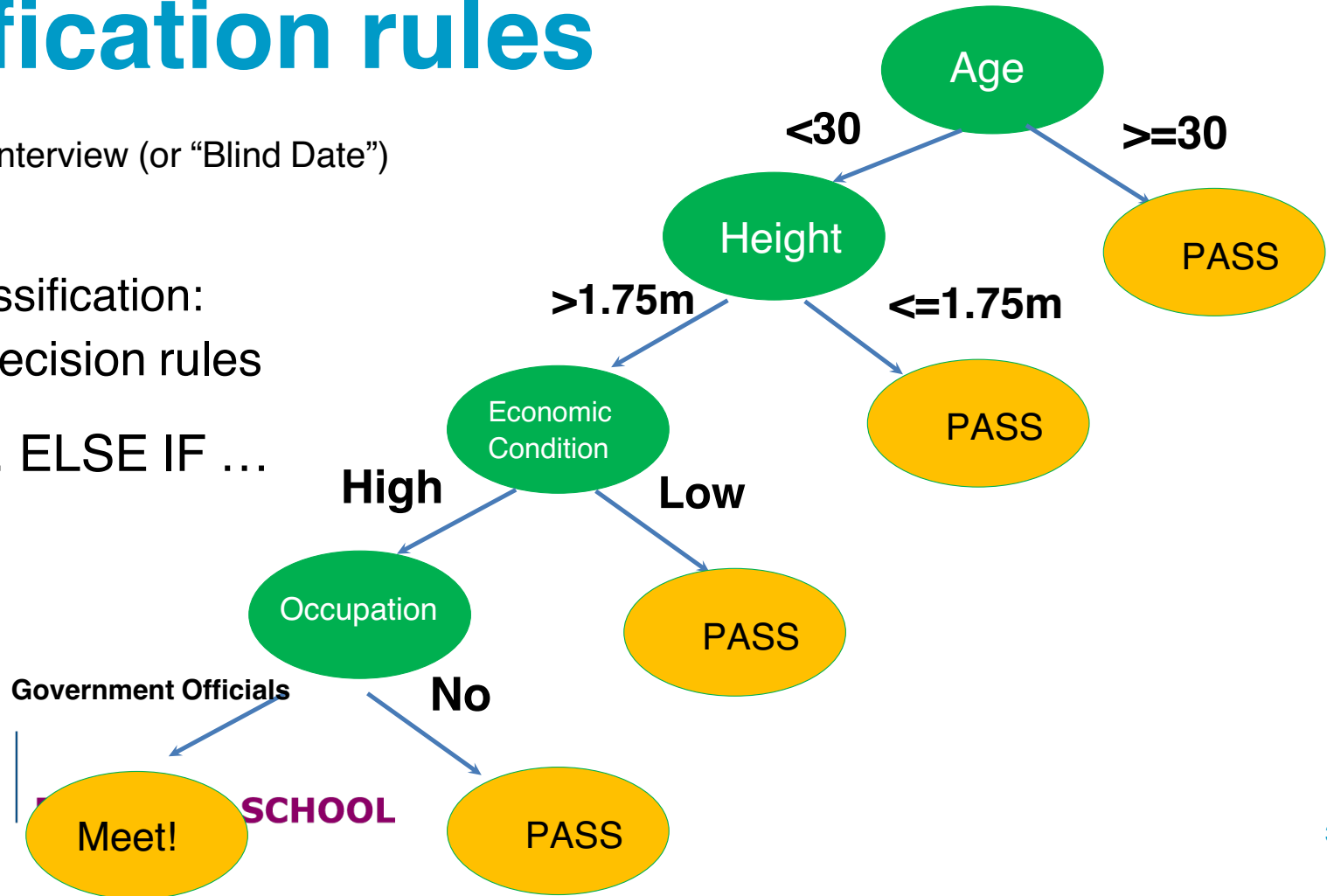
How old is your son?
How tall is he?
How much he earns a month?
How much is the monthly parking
fee in your son's apartment?
Which company is your son
working for?
Does your son have time next Saturday
afternoon to meet my daughter?

Classification rules

Marriage Interview (or “Blind Date”)

Rule-based classification:
classification/decision rules

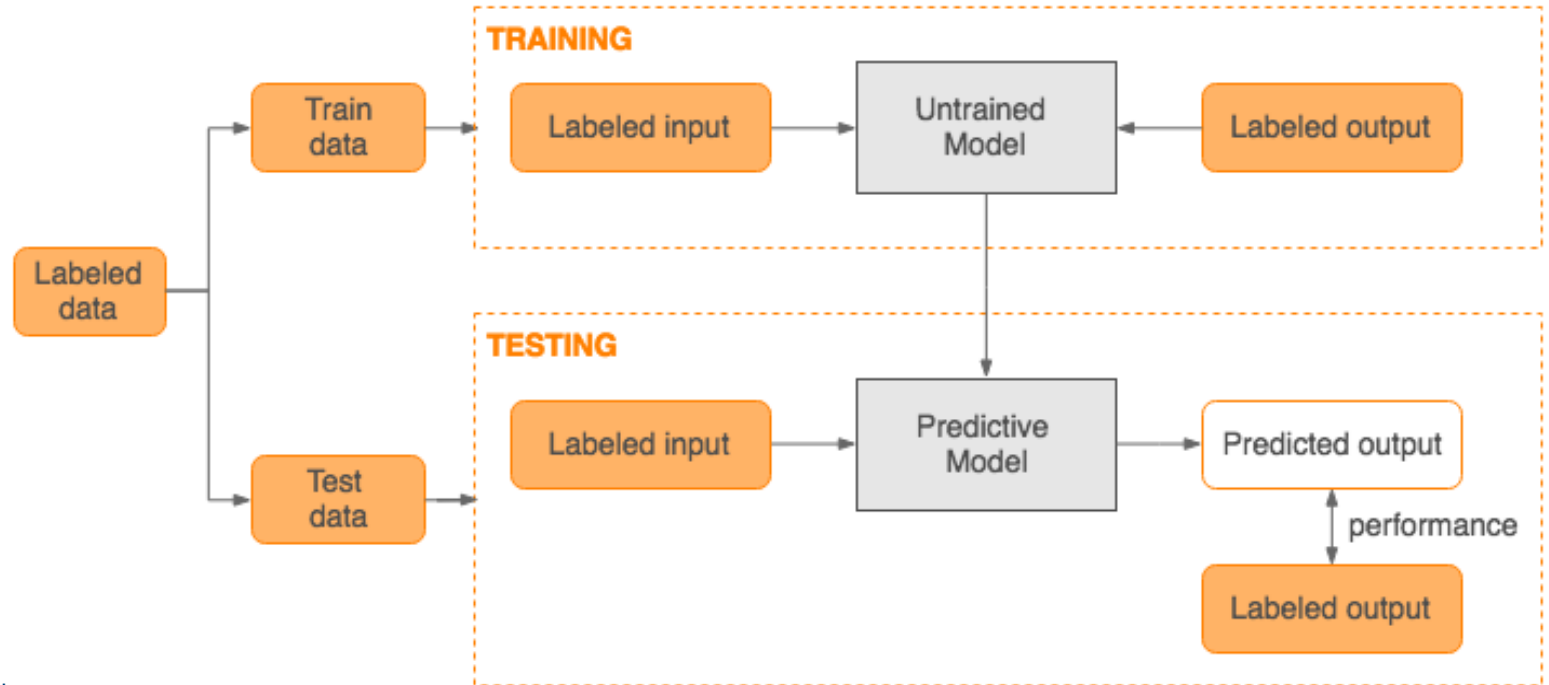
Analogy: IF ... ELSE IF ...
ELSE ...



Train and test data set

After modelling

- Evaluation for performance



Model evaluation

We have been trying to develop machine learning model that has the best performance on the training data set

$\mathbf{X}$	Ground Truth $\mathbf{Y}$	Model Prediction $\hat{\mathbf{Y}}$
$\begin{bmatrix} 4 & \dots & 6 \\ 1 & \dots & 3 \\ \vdots & \vdots & \vdots \\ 7 & \dots & 9 \end{bmatrix}$	$\begin{bmatrix} 435 \\ 598 \\ \vdots \\ 770 \end{bmatrix}$	$\begin{bmatrix} 550 \\ 495 \\ \vdots \\ 760 \end{bmatrix}$
$n \times d$	$n \times 1$	$n \times 1$

Having a good performance on the training data set is not enough!

Model performance

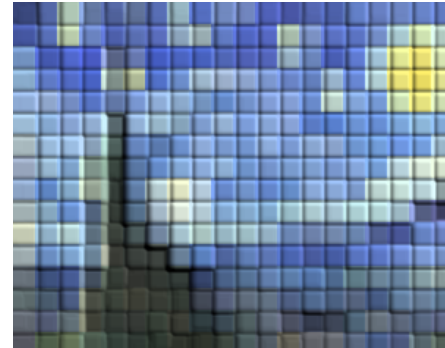
Model Complexity: related to the number of parameters in the model

- ? Complex model: tends to overfit the training dataset
- ? Simple model: tends to underfit the training dataset

Neither of them is good

Very complex model: if you are swamped in the details, you lose the big picture

Very simple model: too crude, not enough pixels



Model performance cont.

You have a training dataset with 20 features and the corresponding binary outcome

Performance on Training Data Set

Model 1. $g(\mathbf{x}) = \beta_0 + \beta_1 x_1 = \beta_0 + \beta_1 x_1 + 0x_2 + 0x_3 + \cdots + 0x_{20}$

Model 2. $g(\mathbf{x}) = \sum_{i=0}^{10} \beta_i x_i = \sum_{i=0}^{10} \beta_i x_i + 0x_{11} + 0x_{12} + \cdots + 0x_{20}$

Model 3. $g(\mathbf{x}) = \sum_{i=0}^{20} \beta_i x_i + \beta_{21} x_1 x_{20} + \beta_{22} \frac{x_3}{x_8} + \beta_{23} x_5^2 + \cdots + \beta_{40} e^{1-\sin(x_{15})}$

$$\min_{\beta} \mathcal{L}(\beta) = \frac{1}{n} \sum_{i=1}^n (\beta^T \mathbf{x}^{(i)} - y^{(i)})^2$$

Model performance cont.

Complexity:

Model 3 > Model 2 > Model 1

If you torture the data harder and longer enough, it will eventually confess, but what it confessed may not be true. Model 3 (complex) is torturing the data severely.

Fundamental Logic: Use algorithm to identify the **general** pattern in the data-generating process from samples (i.e., training dataset) but not any **idiosyncrasy** in samples.

For Model 1: if you are too kind to the data, then why should you expect your data will tell you what it knows?

Intuitively, there should be an optimal level of “torturing”.

Model performance cont.

How should I find this optimal model complexity?

The model complexity is closely related to the number of parameters that need to be estimated in your model

A modification of linear regression: ridge and lasso

Ridge Regression:

$$\min_{\beta} \mathcal{L}_R(\beta) = \underbrace{\frac{1}{n} \sum_{i=1}^n \left(y^{(i)} - \sum_{j=0}^d \beta_j x_j^{(i)} \right)^2}_{\text{Original in-sample error}} + \underbrace{\frac{\lambda}{n} \sum_{j=1}^d \beta_j^2}_{\text{penalty}}$$

Lasso Regression:

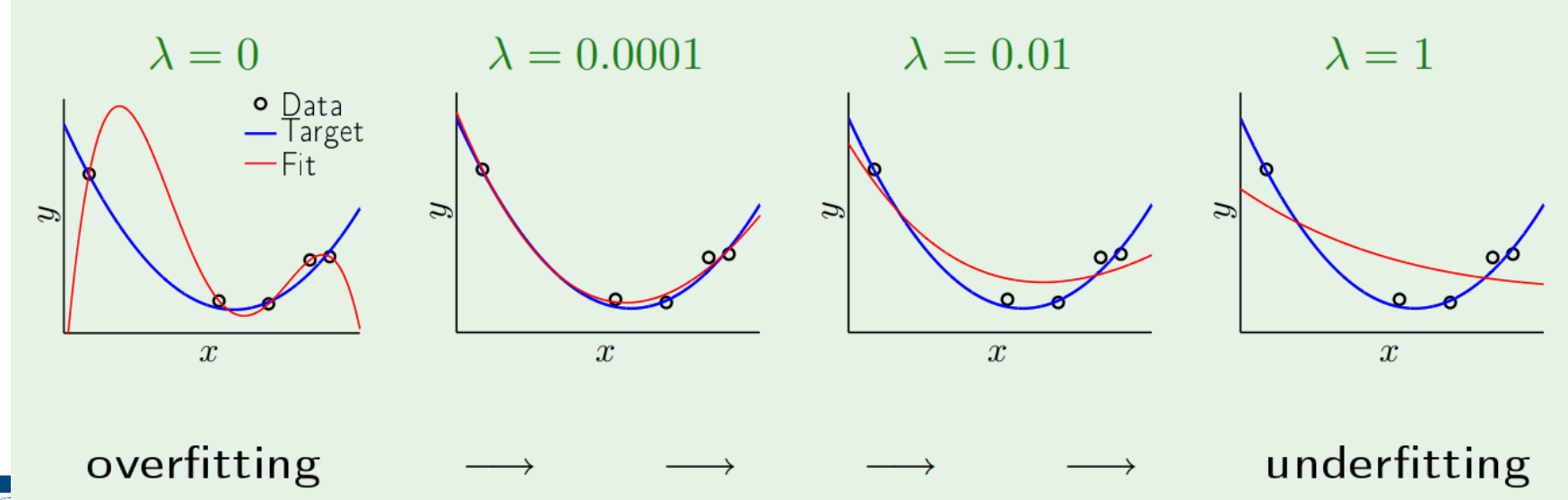
$$\min_{\beta} \mathcal{L}_R(\beta) = \underbrace{\frac{1}{n} \sum_{i=1}^n \left(y^{(i)} - \sum_{j=0}^d \beta_j x_j^{(i)} \right)^2}_{\text{Original in-sample error}} + \underbrace{\frac{\lambda}{n} \sum_{j=1}^d |\beta_j|}_{\text{penalty}}$$



$\lambda \longrightarrow \infty \implies \text{more } \beta_j^* \longrightarrow 0 \implies \text{less complex model}$
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 $\lambda \longrightarrow 0 \implies \text{less } \beta_j^* \longrightarrow 0 \implies \text{more complex model}$ ³⁷

Over- vs. under-fitting

You have 5 samples in the training data set, and you want to use a 5th order polynomial to fit the data



Optimal lambda

General procedure of determining optimal lambda:

1. Have a set of lambdas as candidate, e.g., $\lambda \in \{0.0001, 0.01, 0.1, 1, 10\}$
2. For each fixed λ , run Lasso or Ridge on the training data set, and get $\beta(\lambda)$
3. Using $\beta(\lambda)$ obtained from Step 2, apply it on a test data set
4. Compare all models' performance on the test data set and choose the λ that gives you the best test performance

Train and test data sets

- Parameters vs. hyper-parameters
 - Most algorithms have a set of **hyper-parameter(s)** to control their model complexity
 - Train a set of models with different value of the hyper-parameter(s), then apply these trained models on a test data set and select the model with the best test performance  **Step 6: model evaluation/validation**
 - A test dataset is necessary, but in practice, acquiring extra data can be expensive or even infeasible. How to get an extra independent test dataset?
- Cross validation - the most widely used method to solve this problem

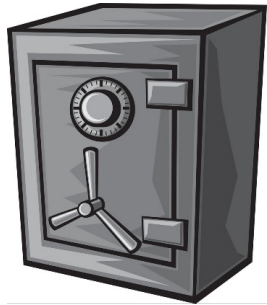
Cross validation

Cross
validation

Leave-one-out (LOO)/ n-fold

K-fold

- Cross validation – a method to test a ML predictive model's performance
 - If extra data is expensive or impossible to get, then I will reuse my training dataset



Training Data Set



“Actual Training” Dataset

Test Dataset

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Performance metrics

- In regression problem, the performance metrics are straightforward and have little ambiguities (e.g., average sum of square errors)
- In classification problem, the situation is less clear
- Many classification problem can be reduced to a Binary Classification problem
 - e.g., credit card applicant default or not, a patient has cancer or not
- The most intuitive metric in the binary classification problem is error rate (or equivalently, accuracy rate)



$$\text{Error Rate} = \frac{\# \text{ of Wrong Predictions}}{\text{Total } \# \text{ of Instances}} = 1 - \text{Accuracy Rate}$$

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But this intuitive metric has several problems

Confusion matrix

- Problems with accuracy rate (or error rate)
 - does not tell the model performance within each class
 - does not address the imbalance class issue

Is there a better metric?

TPR and FPR help us to study the model performance within each class

True Positive Rate

$$= \frac{\text{True Positives}}{\text{All Positives}}$$

$$= \frac{TP}{P}$$

Accuracy rate within the positive class

Model Prediction

	True Class	
	Positive	Negative
Yes ↑	True <i>P</i> ositives	False <i>P</i> ositives
No ↓	False <i>N</i> egatives	True <i>N</i> egatives

False Positive Rate

$$= \frac{\text{False Positives}}{\text{All Negatives}}$$

$$= \frac{FP}{N}$$

Error rate within the negative class

True Positive Rate $\neq$ 1 - False Positive Rate

Accuracy Rate

$$= \frac{\text{True Positives} + \text{True Negatives}}{\text{All Positives} + \text{All Negatives}}$$

$$= \frac{TP + TN}{P + N}$$



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Column Count: P N

Classification measures

Measure	Formula
Classification rate (accuracy)	$\frac{\text{true negative} + \text{true positive}}{\text{total observations}} \times 100$
Misclassification rate	$(1 - \frac{\text{true negative} + \text{true positive}}{\text{total observations}}) \times 100$
Sensitivity (true positive rate)	$\text{Recall} = \frac{TP}{TP + FN}$ $\frac{\text{true positive}}{\text{true positive} + \text{false negative}} \times 100$
Specificity (true negative rate)	$\frac{\text{true negative}}{\text{false positive} + \text{true negative}} \times 100$
1-Specificity (false positive rate)	$\frac{\text{false positive}}{\text{false positive} + \text{true negative}} \times 100$

$$\text{Precision} = \frac{TP}{TP + FP}$$

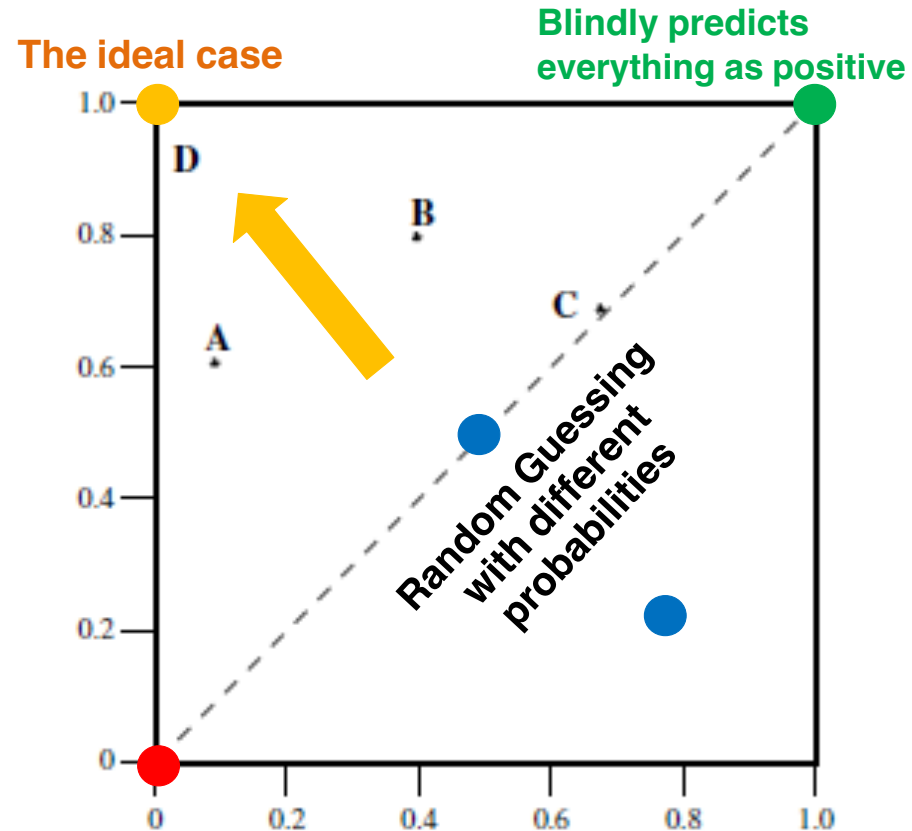
$$\begin{aligned} \text{F1 Score} &= \frac{2}{\frac{1}{\text{Precision}} + \frac{1}{\text{Recall}}} \\ &= \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \end{aligned}$$

Receiver operating characteristics

- Every model for binary classification will generate TPR and FPR from the confusion matrix
- Receiver Operating Characteristics (ROC) helps to visualize the TPR and FPR

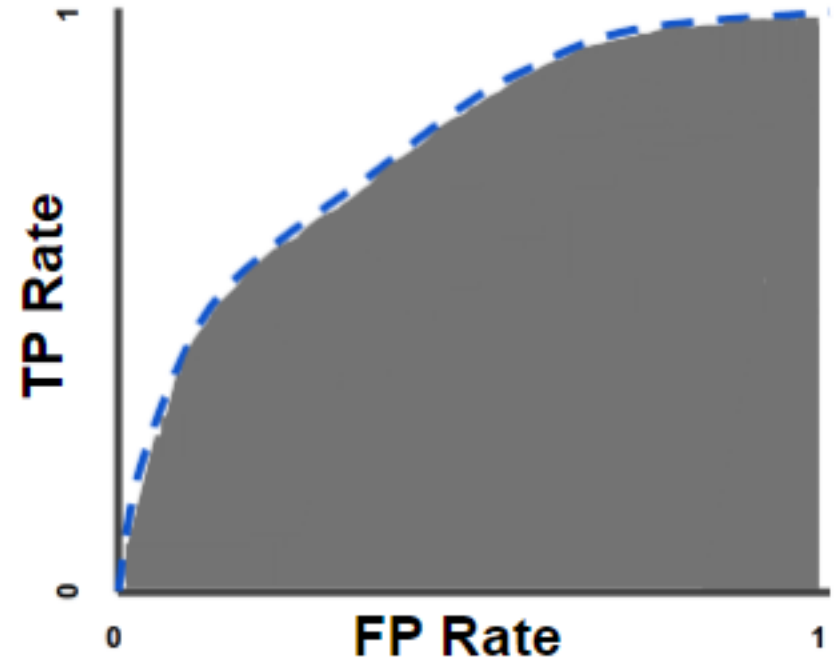
		True Class	
		Positive	Negative
Model Prediction	Yes	True <i>P</i>ositives	False <i>P</i>ositives
	No	False <i>N</i>egatives	True <i>N</i>egatives

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Confusion Matrix



AUC

- Area Under the ROC Curve (AUC)
 - Measures the entire two-dimensional area underneath the entire ROC curve (think integral calculus) from (0,0) to (1,1)
 - Provides an aggregate measure of performance across all possible classification thresholds
 - AUC ranges in value from 0 to 1
 - A model whose predictions are 100% wrong has an AUC of 0.0; one whose predictions are 100% correct has an AUC of 1.0



Summary of ROC and AUC

1. Error rate is often not enough to evaluate the performance of models. We need to know the error within each subclass, which leads us to the **TPR** and **FPR**.
2. A model with high TPR and low FPR is considered as good. Ideally, we want $TPR=1$ and $FPR = 0$. But there is a tradeoff between TPR and FPR
 - One classifier is better than another if its corresponding point on the ROC Graph is to the 'north-west' of the other's
3. By setting different values of the cutoff probability, we can generate ROC curve from our model
4. We can use AUC (area under the ROC curve) as the performance metric to select the best model. The more AUC is close to 1, the better

Unsupervised learning

- Recap:
 - Data: unlabelled data
 - Discover patterns; no idea whether they are right or wrong
- Clustering
 - Group similar objects together
 - Separate dissimilar objects into other groups
- Usage
 - A standalone tool to get insights into data distribution
 - A pre-processing step for other models/algorithms

When to use clustering

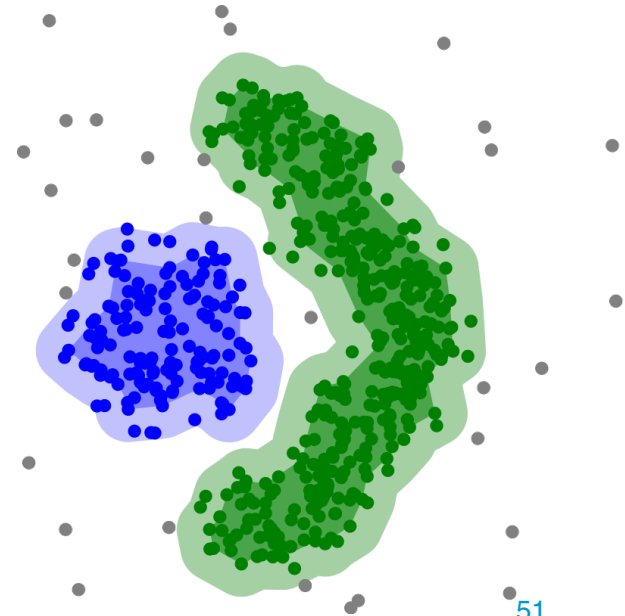
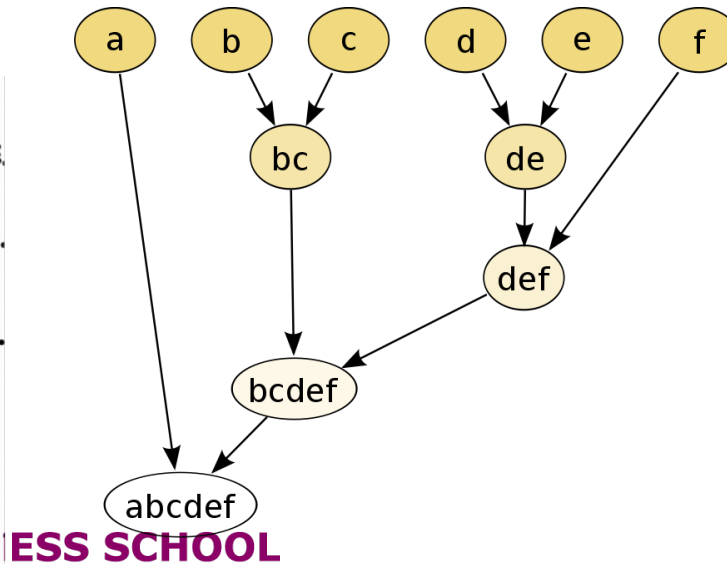
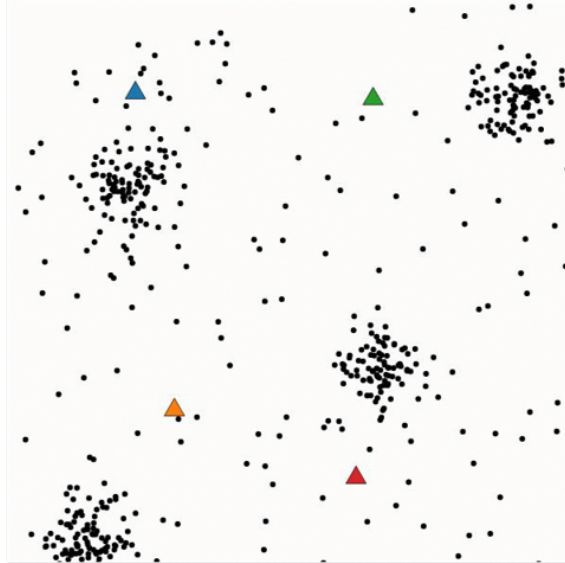
- Homogeneity
 - Closer to ones in the same cluster vs different clusters
- Separation
 - Clear distinction, few boundary cases
- Size/Shape
 - Similar size (count) and shape (spherical or ellipsoid)
- Interpretability
 - Real-world relevance based on attributes

Cluster quality evaluation

- Internal Metrics
 - Cohesion vs Separation: Score for each value of K
- External Metrics
 - Alignment with known categories (related domains)
- Manual Evaluation
 - Expert opinion: visual and subjective interpretation
- Indirect Evaluation
 - Utility of clusters in accomplishing the business objectives

Types of clustering

- Partitioning
- Hierarchical
- Density-based



Comparison

Partitioning	Hierarchical	Density-based
Generates non-overlapping clusters	Generates a tree-like structure of clusters	Generates clusters based on data density
Easy, fast and suitable for large datasets	Captures natural hierarchy within clusters	Identifies clusters of arbitrary shapes
Deciding the number of clusters is an art	Too slow for medium to large datasets	Suitable for specific density patterns only

K-means

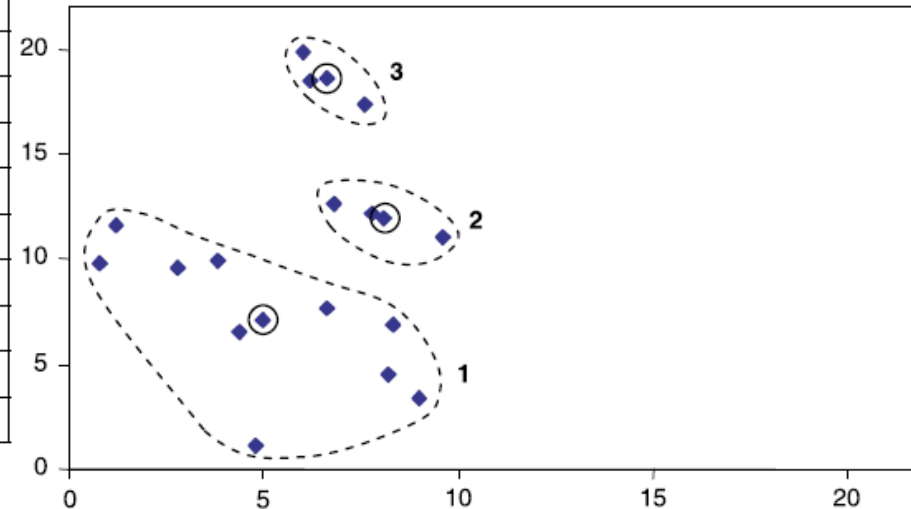
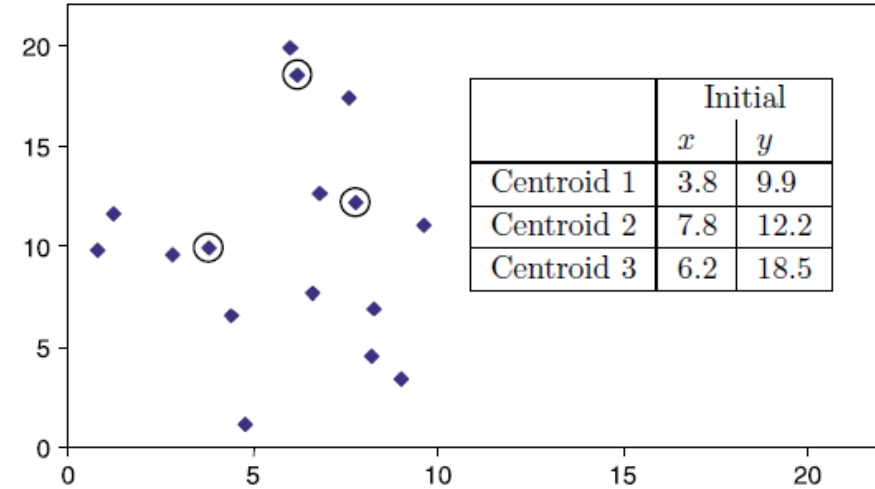
- How to determine similarity / dissimilarity?
 - Distance measures between objects (cosine similarity, Euclidean distance, ...)
 - Distance measures between clusters (average/single/complete linkage)
- Distance-based method for partition
 - K-means
 - K: predefine the number of clusters we want to form
 - Each cluster is represented by the centre of the cluster

1. Choose a value of k .
2. Select k objects in an arbitrary fashion. Use these as the initial set of k centroids.
3. Assign each of the objects to the cluster for which it is nearest to the centroid.
4. Recalculate the centroids of the k clusters.
5. Repeat steps 3 and 4 until the centroids no longer move.

K-means algorithm

- 16 objects, 2 attributes
- $K=3$
- Calculate distances to each centre
- Recalculate centres
- Repeat until convergence

x	y
6.8	12.6
0.8	9.8
1.2	11.6
2.8	9.6
3.8	9.9
4.4	6.5
4.8	1.1
6.0	19.9
6.2	18.5
7.6	17.4
7.8	12.2
6.6	7.7
8.2	4.5
8.4	6.9
9.0	3.4
9.6	11.1



Further questions on k-means

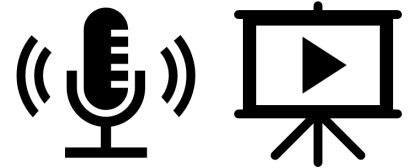
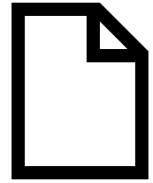
- How to decide the optimal k?
- How to evaluate the quality of a set of clusters?
- One simple method:
 - Let's measure the quality of a set of clusters using the value of an *objective function* which we will take to be **the sum of the squares of the distances of each point from the centroid of the cluster to which it is assigned (WSSC)**
 - Try different k values
 - Chose the set of clusters with the smallest objective function
- How to choose the initial centres?
- One simple method:
 - Random selection

More advanced questions:

- Hard vs. soft
 - Exclusive membership
- Outliers
 - OK vs. strict
- Overlap: OK vs. NO

Unstructured data

- So far, we have been working with processed structured data (structure/physical composition, order (or sequencing), metadata)
- Documents
 - Text extraction & natural language processing
- Images
 - Resize, rotate, crop
 - Pixel data (RGB or Grayscale), labelling
- Audio & Video
 - Frequency analysis, frame extraction
 - Transcription, object detection, labelling



Text mining

- Recall last week's example on "Spam Text Classification"
- Representation
 - Bag-of-words: unordered collection (i.e., bags) of words, keeping frequency but disregarding grammar and word order
 - N-grams: a sequence of N words
 - E.g., "free account" is a 2-gram (bigram)
 - Word embeddings (e.g., Word2Vec): create a distributed representation of words into numeric vectors
 - Tf-idf (term frequency-inverse document frequency): reflects the importance of a word in a document, relative to its frequency across all documents (term-weighting)
 - Topic models: represent document by a probability distribution over topics (topics are then represented by a distribution over words)

Text mining cont.

- Stop words:
 - A list of common words that are likely to be useless for classification
 - Remove their occurrences before representation
- Stemming
 - Many words have variants based on same linguistic roots
 - Treat words with the same root equivalently
- Text mining applications:
 - Named entity recognition, sentiment analyses and opinion mining, topic modelling, information extraction, information summaries ...

Step 4: data transformation



Streaming data

- So far, we have been working with static data
- Streaming data
 - Data in motion or real-time data
 - Continuous flow: generate & transmit vs Store
 - Sources: sensors, IoT, trading, social media
 - Time sensitive: relevant for quick decisions
 - Size vs volume: A few KB/MB every few seconds

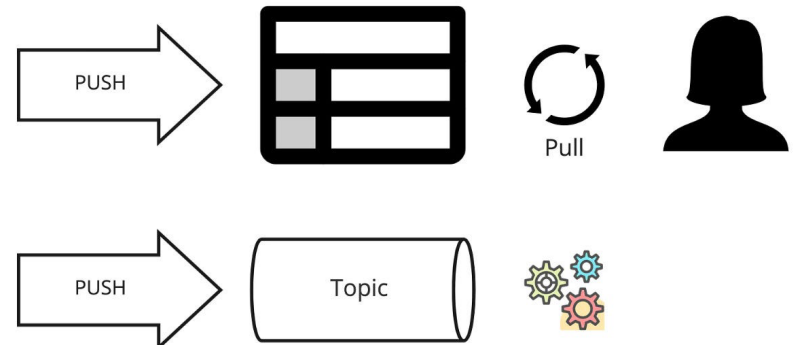
ML with streaming data

- Process data within a rolling time window
- Handle a few records at a time
- Quick response functions (sub-seconds)
- Needs transient storage and processing layer
- Continuous learning with real-time feedback

Example in Python

- Data collection
- Pull vs. push
- Real-time data (request/response pull basis)
 - Package: requests
- Streaming API (streaming push)
 - Package: sseclient

Step 2: data understanding



Future trends

- No-code, Low-Code ML
 - Workflow approach, Instruction based
- Auto-ML
 - Goal-driven selection of method, algorithm and parameters
- Self-Supervised Learning
 - Generate labels or targets without guidance
- Generative Models
 - Access to real-time data; Bias correction

Summary

This week:

- Regression – Step 5 & 6
- Classification – Step 5 & 6
- Model evaluation – Step 6
- Clustering – Step 5 & 6